

Sheetmetal Forming

Sheet metal parts are made by applying cutting and deformation shaping processes to stock sheet metal. Typically, sheet metal comes in thicknesses of maximum 6mm, and its mechanical properties are influenced by the grain direction of the metal. Generally, sheet metal is strong in tensile loading, however easily buckles under compression and bends/twists easily.

Different materials can be used for making sheet metal parts, but zinc, steel and aluminium alloys are most commonly used, in different thicknesses depending on the application.

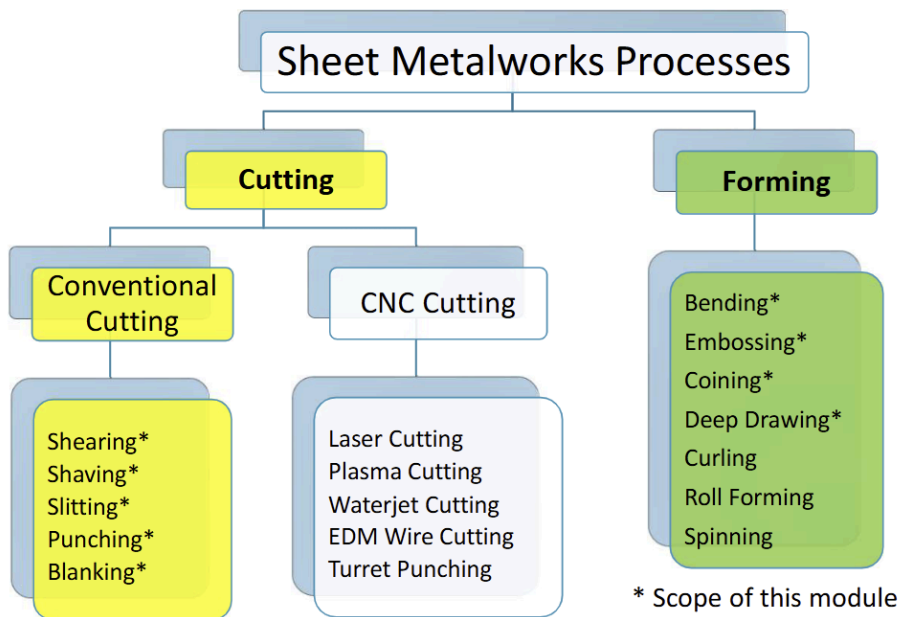
Common sheetmetal materials (generic):

- Pre-coated Cold Rolled Steel (CRS)
 - Electro-Galvanized (EG)
 - Hot Dip Zinc Coated
- Aluminium Alloy
- Stainless Steel
- Beryllium Copper
- Tin plated steel
- and more...

Commonly available thickness	
EG / Hot Dip Zinc Coated CRS (mm)	Stainless Steel 301/304 (mm)
0.4	0.2
0.5	0.3
0.6	0.4
0.8	0.5
0.9	0.6
1.0	0.8
1.2	1.0
1.5	1.1
1.6	1.2
2.0	1.3
	1.4
	1.5
	1.6

Sheetmetal Processes

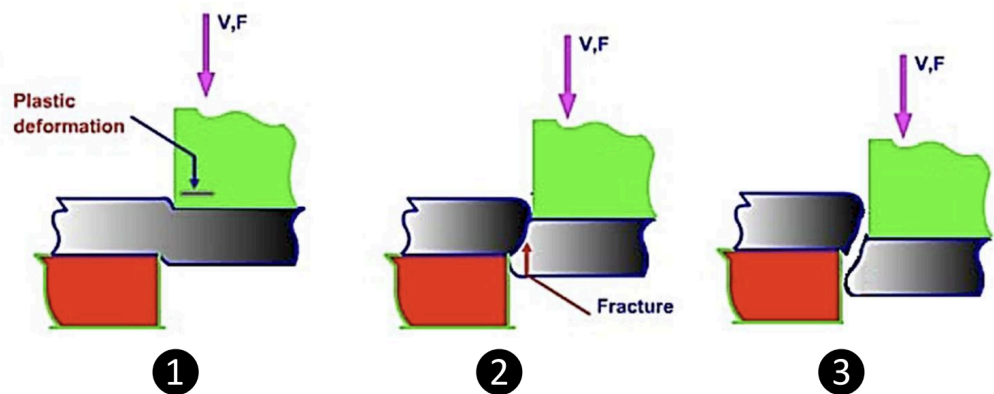
To process sheet metal into the desired part, there are different kinds of operations that can be used:



Cutting

Shearing

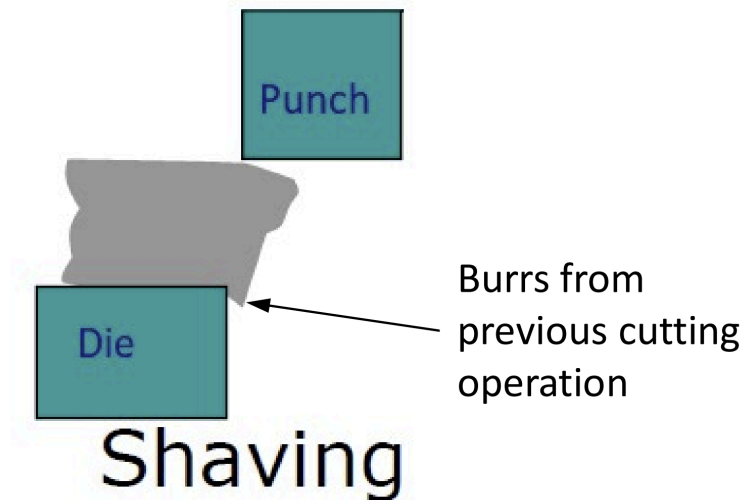
Shearing involves separating the metal fully across the bar, by applying a high vertical force between two cutting edges, to cut it into smaller pieces. This process involves plastic deformation of the sheet metal, followed by fracturing the metal, and then shearing:



This causes a burr to form opposite to the cutting direction, in this case at the bottom.

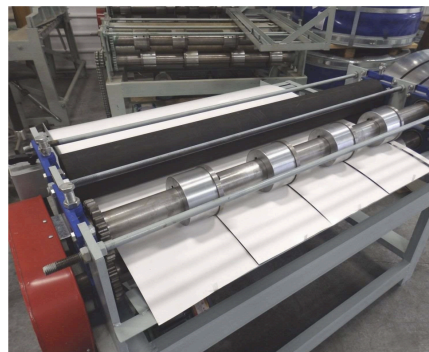
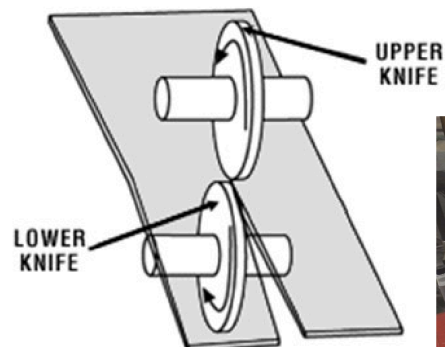
Shaving

Shaving is a finishing operation that removes burrs from sheared edges, by using a punch to shave off the burr, allowing for smoother finish and better dimensional accuracy.



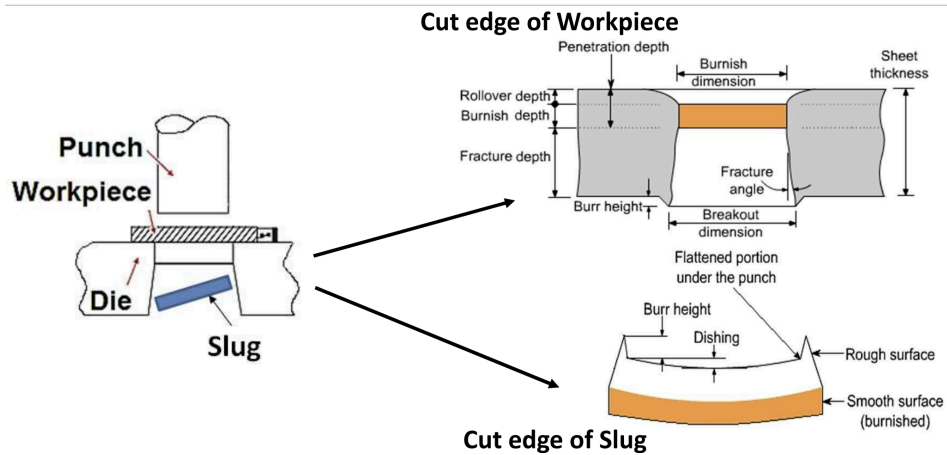
Slitting

Involves using two roller cutting edges (slitters) to cut a wide coil of metal into narrower coils of metal, by pushing the metal into the slitters



Punching & Blanking

Both punching and blanking operate on the same principle, by cutting sheet metal along a closed contour, using a punch (cutting tool) to force the sheet through a die (opening block). As the punch has to fit through the die, they must have the same cross section with some clearance to properly punch out the metal. The clearance can be controlled to control the cutting force & accuracy, and the edge geometry.



The difference between punching and blanking is that punching cuts out the unwanted material, like holes, while blanking is for cutting the desired part out of the stock sheet metal.

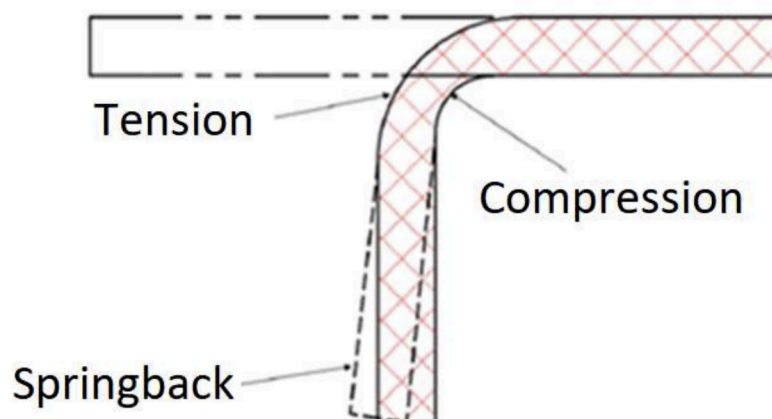
For example, to make washers, punching is first done to remove the desired hole, followed by blanking to cut the circular contour.

Forming

These processes involve deforming the metal past the elastic limit, to obtain the desired shape.

Bending

During bending, tensile and compressive stresses are setup in the bend, about the neutral axis. When bending a part, spring back may occur, which is when the metal bends back slightly. As such, over bending is needed to compensate for the spring back to get the desired angle.



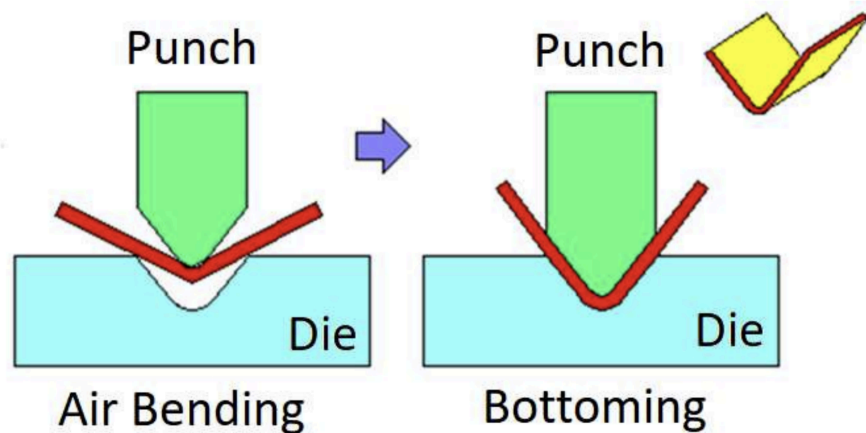
To perform bending, there are some common operations:

V Bending

The sheet metal bends along a straight line with a V shaped punch and die. This setup allows for two ways to achieve a certain bend

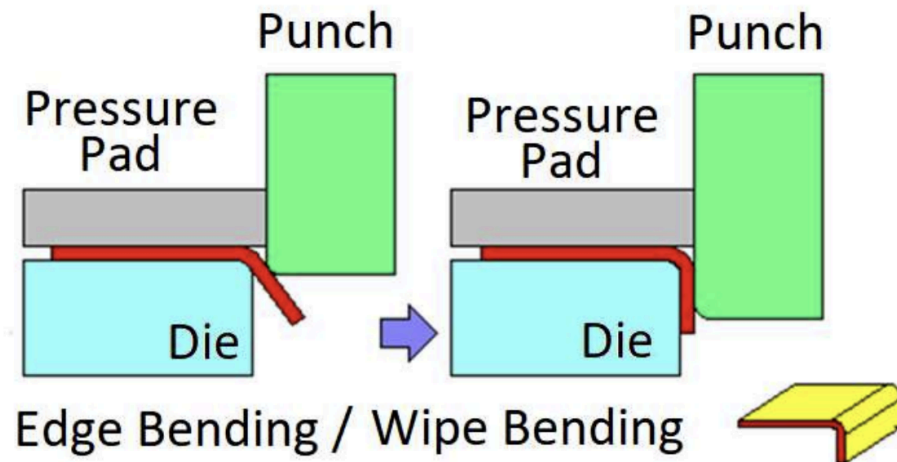
Air Bending, is when the sheet metal is not pressed fully up to the end of the die, hence less bending force is needed but there will be more variation/lower accuracy in the bend angle.

Bottoming is when the sheet metal is pressed fully up to the end of the die, requiring more bending force but a more accurate part/more consistent bend angle



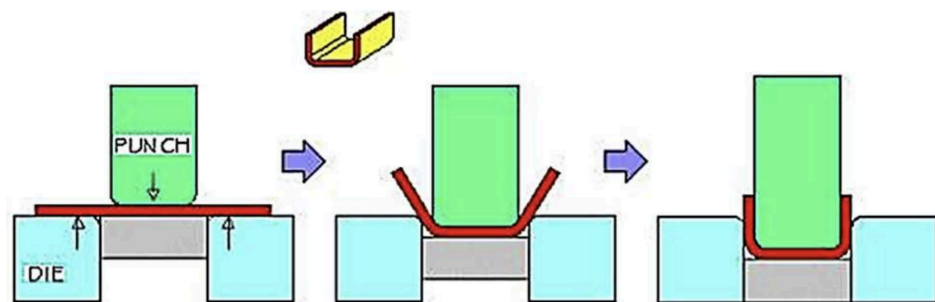
Edge Bend/Wipe Bend

The part is placed on a die in a cantilever manner and a punch pressed on the unsupported part to achieve bends at the ends of sheet metal



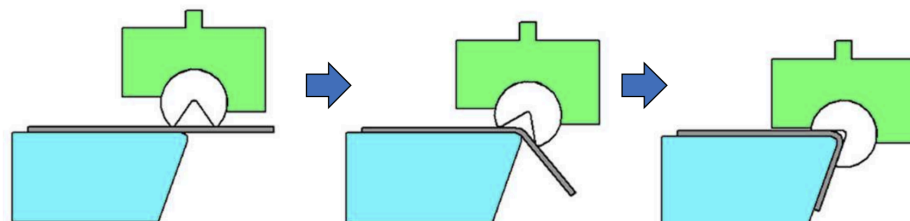
U Bending

U Bending works in a similar manner to V bending, but the die has a movable portion that moves with the punch to form a flat base for the bend.



Rotary Bending

Rotary Bending has a V shaped die and the punch has a V shaped notch that can swivel, allowing bends of more than 180 degrees:



Embossing

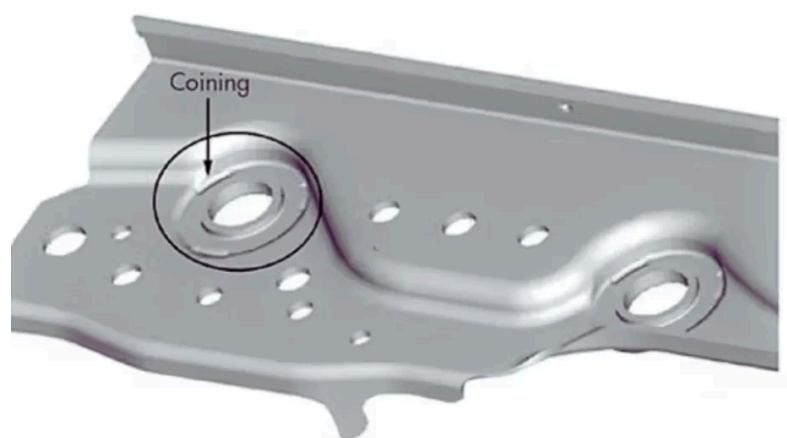
This method creates indentations via a pair of matching contours in the die and punch. This causes the metal to bulge out in the specific

pattern, causing stretching and thinning at the edges of the pattern, but without cutting.

This method is most useful for marking logos, strengthening parts or creating textures surfaces.

Coining

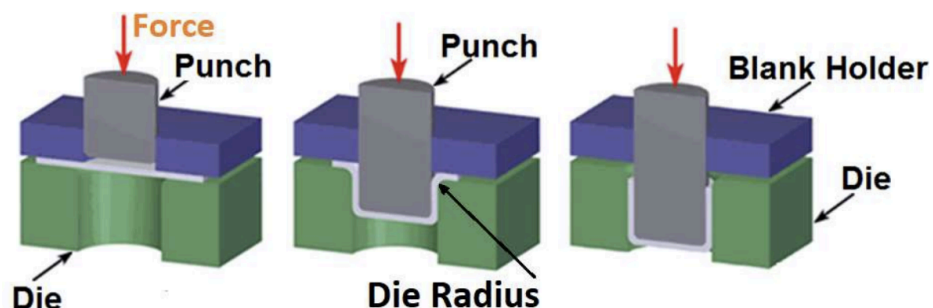
Coining is similar to embossing, except the punch and dies do not need to have the same contours. With this method, finer and detailed surface finish can be achieved, and can be applied for the same purposes as embossing



However, this method involves high tonnage required by the machine, has tooling limitations and causes a higher machine wear rate.

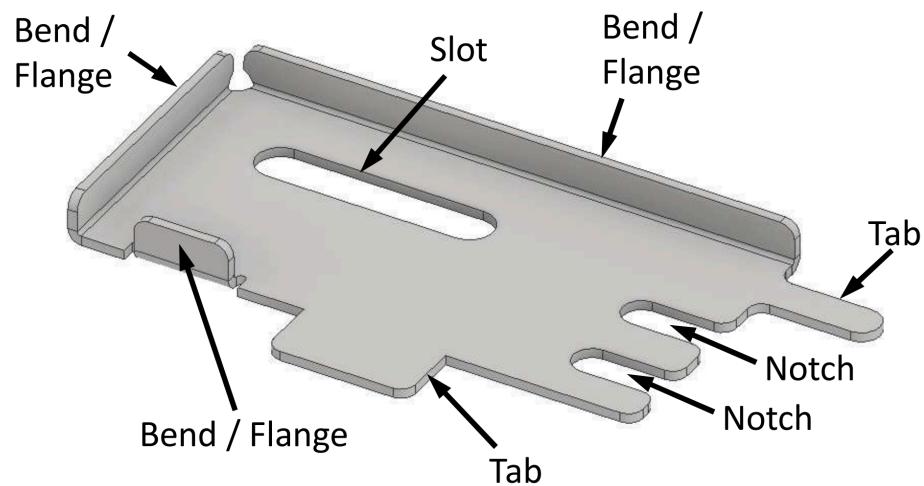
Deep Drawing

This method is similar to U-Bending, except the whole sheet metal is fit into the die, causing the whole part to form a revolved U bend, like a cup, useful for kitchenware, sinks and cups.



Sheetmetal Part Features

Shown are some primary sheet metal part features:

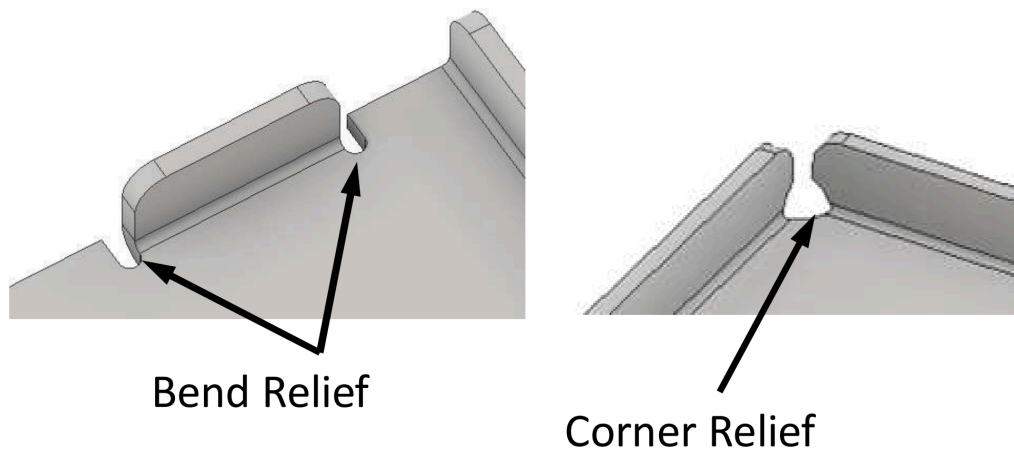


Bend/Flange: any bend/fold along an edge

Tab: Any protrusion from the main body

Notch: A narrow cut out from the edge

There are also features meant for better part life/easier manufacture:

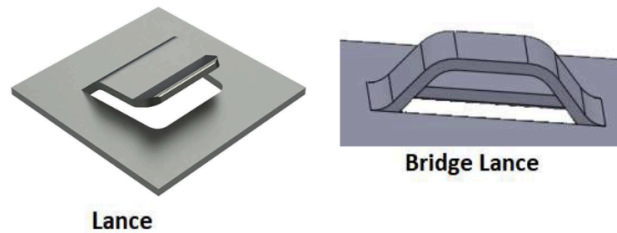


Bend Relief: A cut out at the root of the bend to avoid tearing when bending

Corner Relief: A cut out at the root of a corner, to avoid tearing and allow flow of material

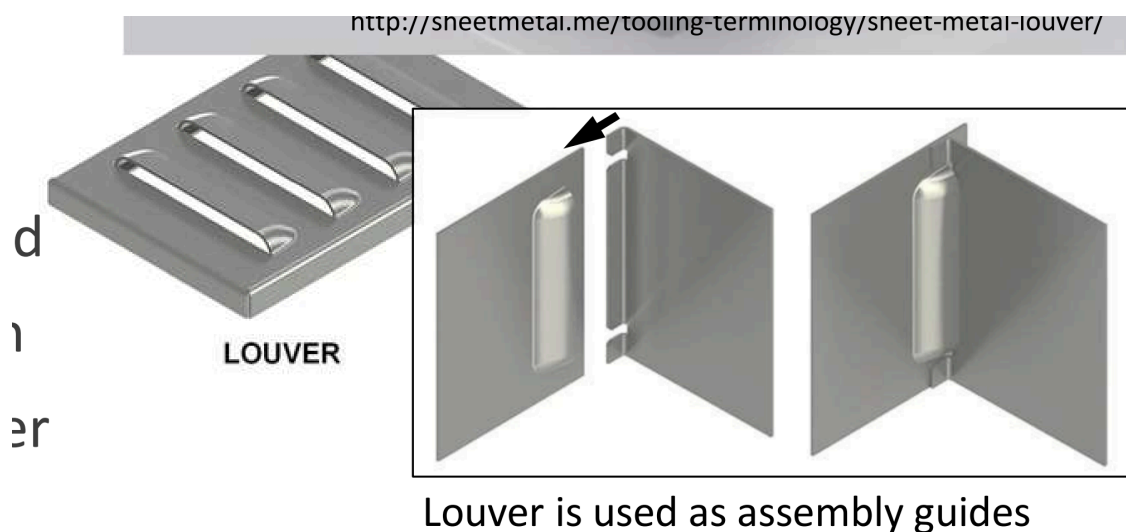
Some other secondary sheet metal features include:

Lances: a sheared and bend section due a strike of a die, causing only a change in geometry

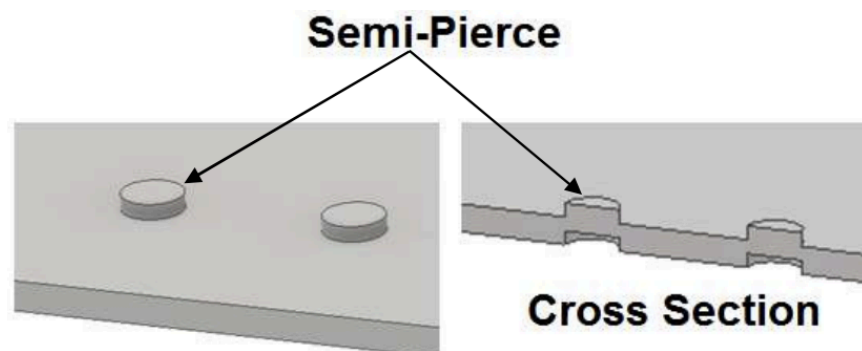


Emboss: covered earlier, indentations due to two similar contour dies

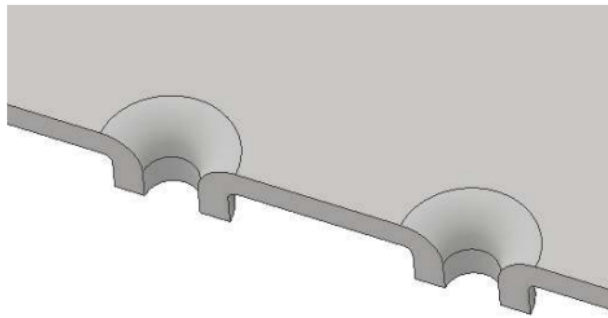
Louvers: similar to a lance, however slit on one edge and pressed into a die to create an opening along one edge, for ventilation or assembly guides as a slot for flanges



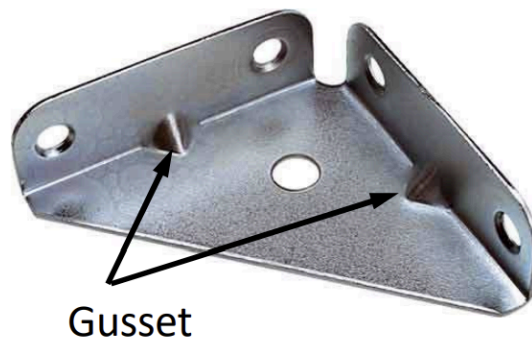
Semi Piecing: piercing/half shearing the metal without tearing, similar to embossing, max depth $0.75t$, diameter $2t$.



Extruded Holes: metal surrounding hole is extruded beyond sheet, usually for self tapping screws



Gussets: Like a rib, provides support, strength and stiffness



Hems: forms rounded edge of sheet metal end by folding it onto itself, for aesthetics, strength and safety. comes in 4 forms, closed, open, round and teardrop.



Design Considerations

Sheetmetal Tolerances

There are two common ways to manufacture the sheet metal part prior to bending, and their tolerances for the different possible features and dimensions are shown:

Features	Stamping Machine Tolerance (mm)	Laser Cut & Press Brake (mm)
Hole Size	± 0.08	± 0.15
Feature to Feature Distance (Cut Edge to Cut Edge)	± 0.1	± 0.15
Bend to Bend Distance	± 0.3	± 0.4
Feature to Bend Distance (Cut edge to Bend Distance)	± 0.25	± 0.3
Bend Angle	$\pm 1^\circ$	$\pm 1.5^\circ$

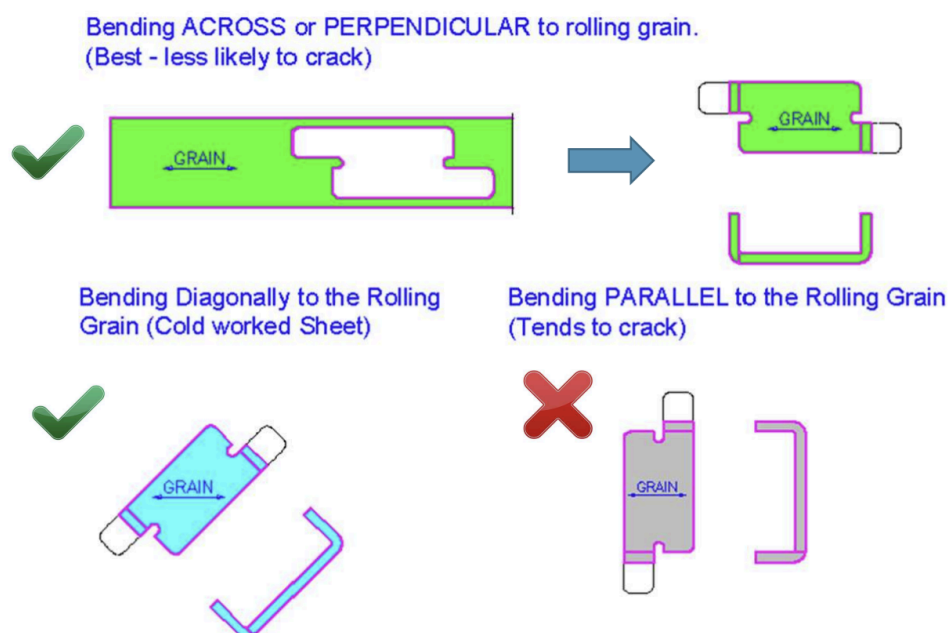
Bending Considerations

Generally, there are three factors to consider when making bends on sheet metal to avoid cracking on the outer surface of the bend.

Grain Direction

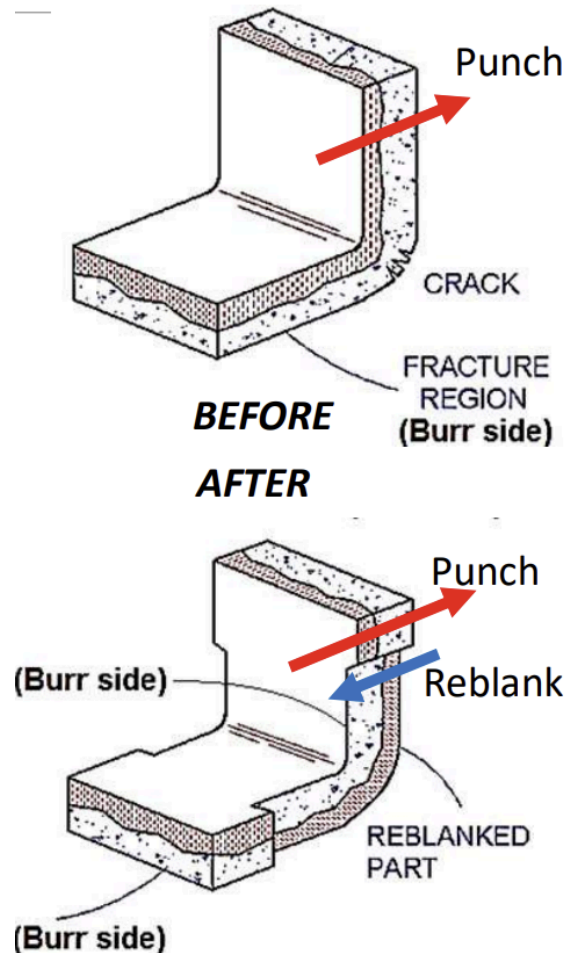
when sheet metal is made by cold rolling into flat sheets, the grains are distorted in one direction, causing a certain grain direction in the sheets.

Bends should not be parallel to the grain direction otherwise cracking becomes more prone, and instead should be minimum 45 degrees, or best perpendicular to the grain direction.



Cut Edge Quality

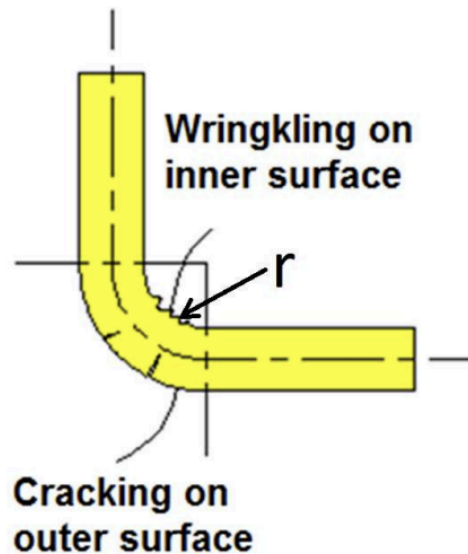
As bending puts tension on the outer side, burrs may form which may encourage fracture. Re-blanking can be done to deburr the outer side, which makes it less likely for fracture.



Bend Radius

Bend radius is always measured as the internal bend radius, r .

Too small of a bend radius causing cracking externally (said earlier) and wrinkling internally:



Hence, the minimum bend radius depends on material thickness, material stiffness, bend angle and grain direction.

Higher thickness, stiffness, bend angle, and more aligned grain direction to bends increase the minimum bend radius.

Typically, bend radius is proportional to the thickness t .

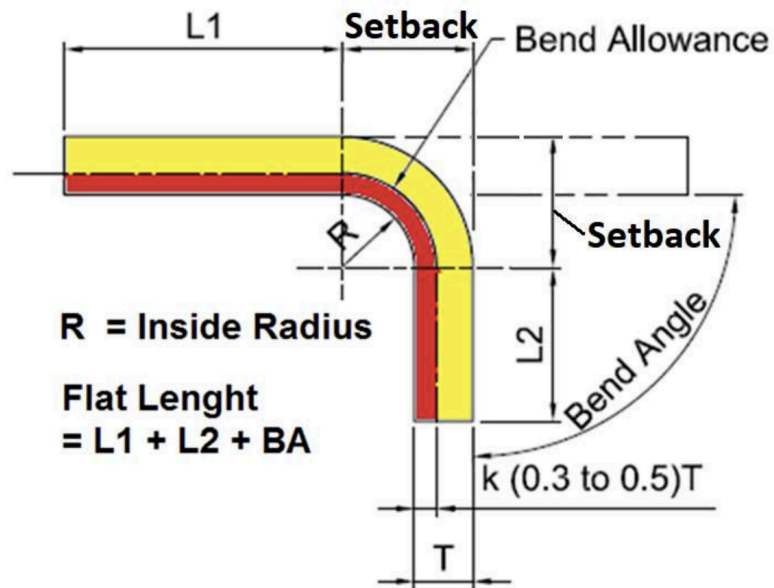
Material	Condition	
	Soft	Hard
Aluminium Alloys	0	6T
Beryllium Copper	0	4T
Brass (Low-leaded)	0	2T
Magnesium	5T	13T
Steel, Austenitic Stainless	0.5T	6T
Steel, Low Carbon	0.5T	4T
Titanium	0.7T	3T
Titanium Alloys	2.6T	4T

Bend Geometry

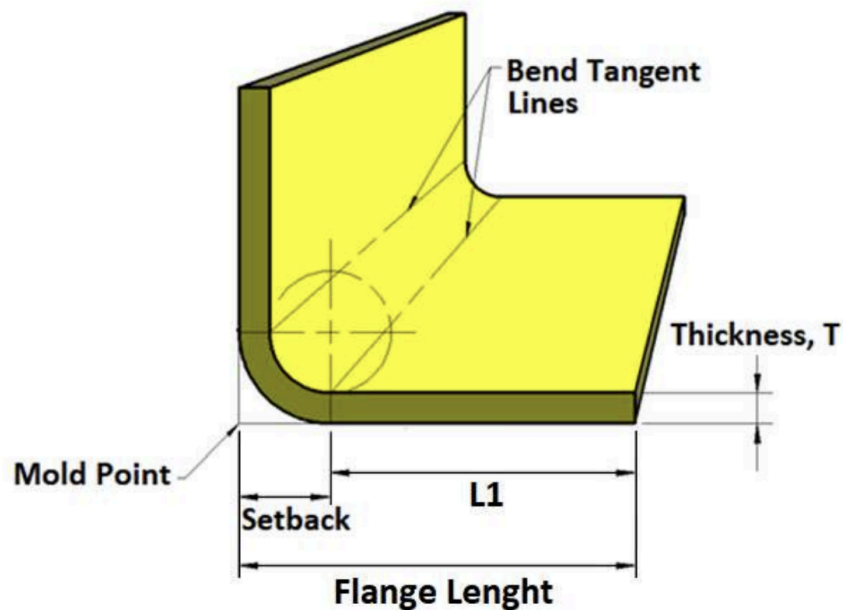
During bending, the neutral axis shifts closer to the internal bend. Bend allowance is the length of this neutral axis around the bend, which represents the extra length of metal needed to make the bend. Bend allowance is given as

$$BA = \frac{\theta^\circ}{360} \times 2\pi(R + kT)$$

where R is the internal radius, θ is the bend angle from rest position, and k is an experimental value that ranges from 0.3 to 0.5. $k=0.5$ means that the neutral axis is in the middle of the thickness.



To find the total flattened length, it is the sum of the tab/flanges + BA. The total flattened length is also called the development length

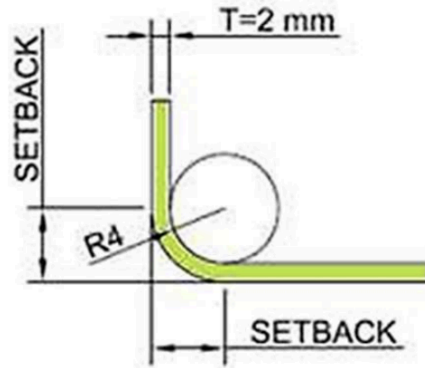


Other features include the mold point; the intersection of the extension of the flanges, the bend tangent lines; the start and end of

the bend, and the setback; the distance between the mold point and bend tangent line.

To find setback, if the bend is 90 degrees,

$$\text{Setback} = R + T$$



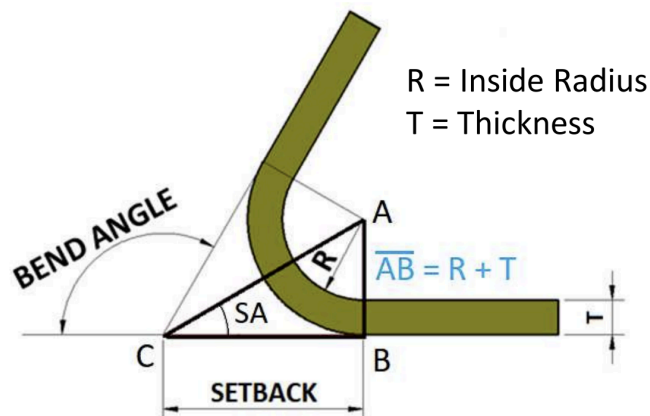
But if the bend is not 90 degrees,

The setback angle is

$$\theta_{sa} = \frac{1}{2}(180 - \theta)$$

and the setback is

$$\text{Setback} = \frac{R + T}{\tan(\theta_{sa})}$$



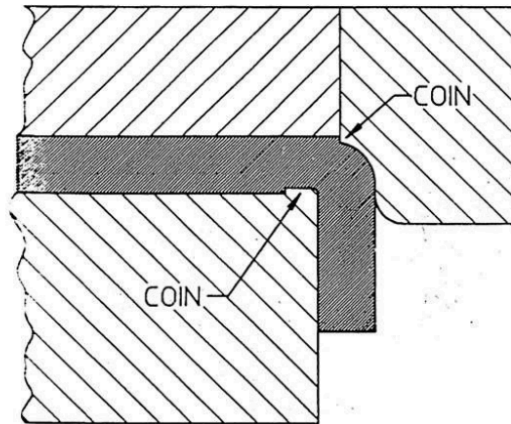
Spring Back

Spring back is when the metal unbends slightly due to its slight elasticity; the compressed site extends slightly while the stretched

side contracts slightly.

Hence, bend angle tolerances are made, from ± 1 to ± 3 degrees.

spring back can be mitigated by overbending, or by coining the edge to set the corner (corner setting).



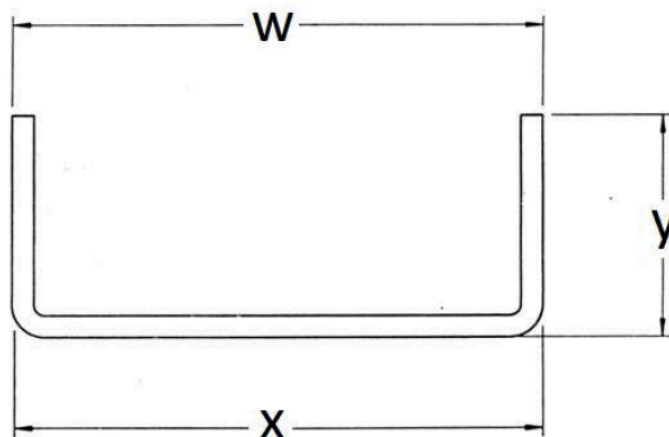
Corner Setting

spring back is affected by the thickness, material stiffness, radius, bend angle, and the part's shape

Sheetmetal Design Principles

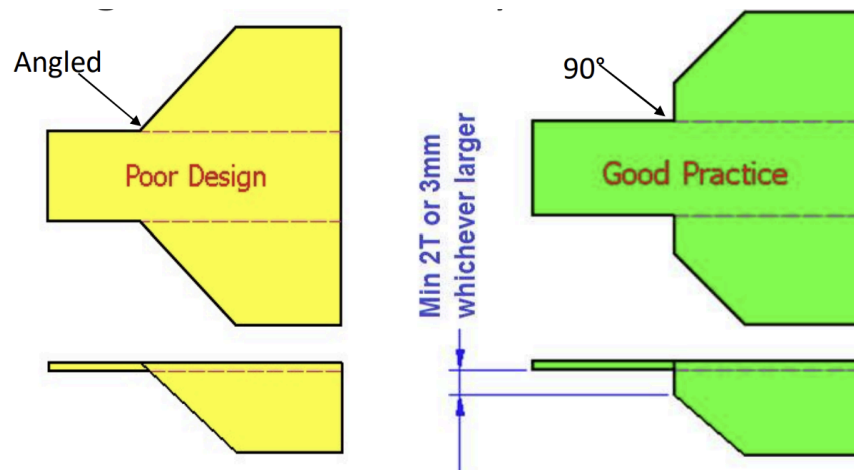
Dimensioning

bounding dimensions should be used in most cases. for embosses, lances, etc., the depth should be measured from the face it is displaced from, unless the total height is critical.



Bending

bends should be made at 90 degree tabs, otherwise buckling may occur.



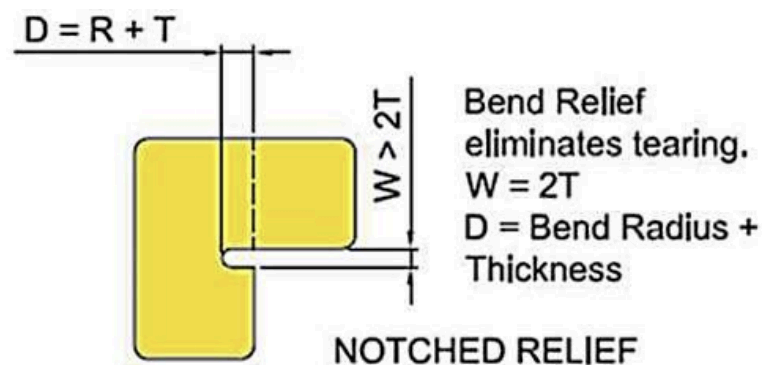
Flanges & Hole/Slot Location

min length $2.5t+r$ for flanges, and hence min distance of hole/slot to bend is same as $2.5t+r$

Bend Relief

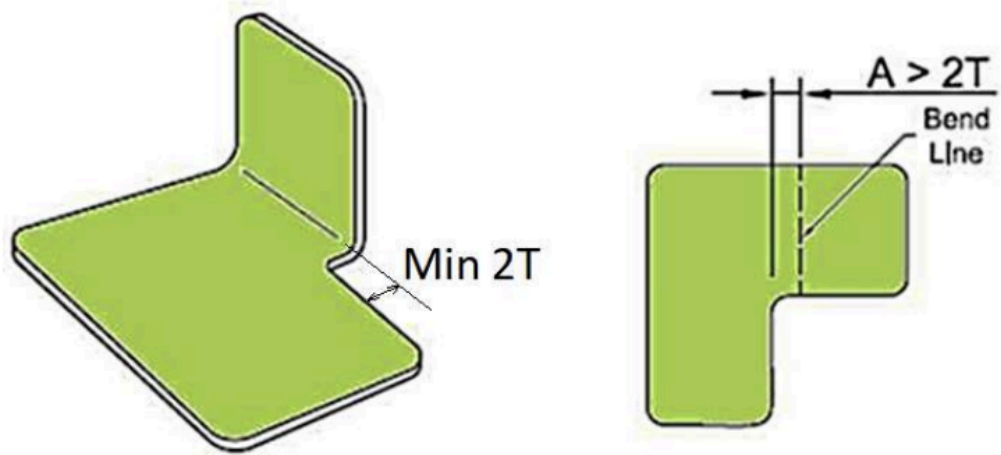
Allows clearance and prevents tearing. Use wide enough and rectangular or oblong relief slots, not triangular due to stress concentration.

notch relief: width = $2t$, depth = $r+t$ minimum.



or bend reliefs can be added by cutting out circles of radius min $1.5r$.

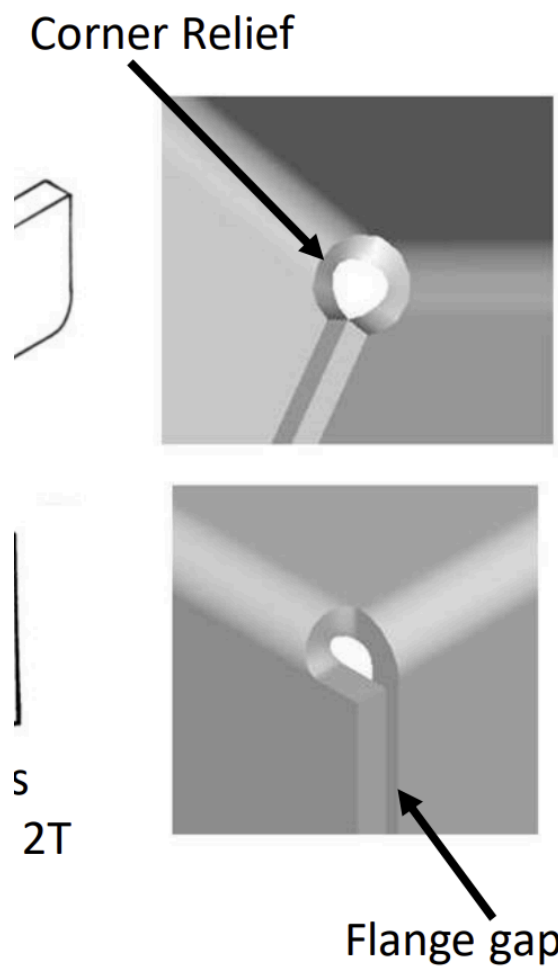
alternatively, a small tab extension can be placed before the bend to mitigate notches entirely, with minimum tab extension $2t$.



Corner Relief & Flange Gap

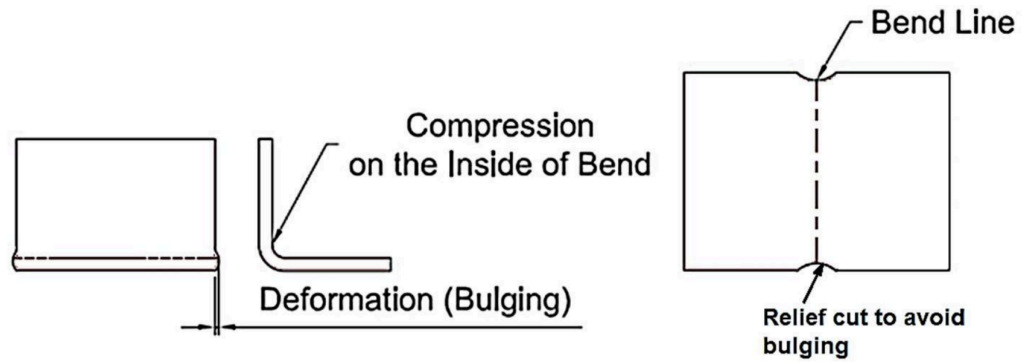
similar to bend relief, reduces bulging and allows material flow. cutout of circle min radius $2t$ is used.

a flange gap may be used for corners like so:

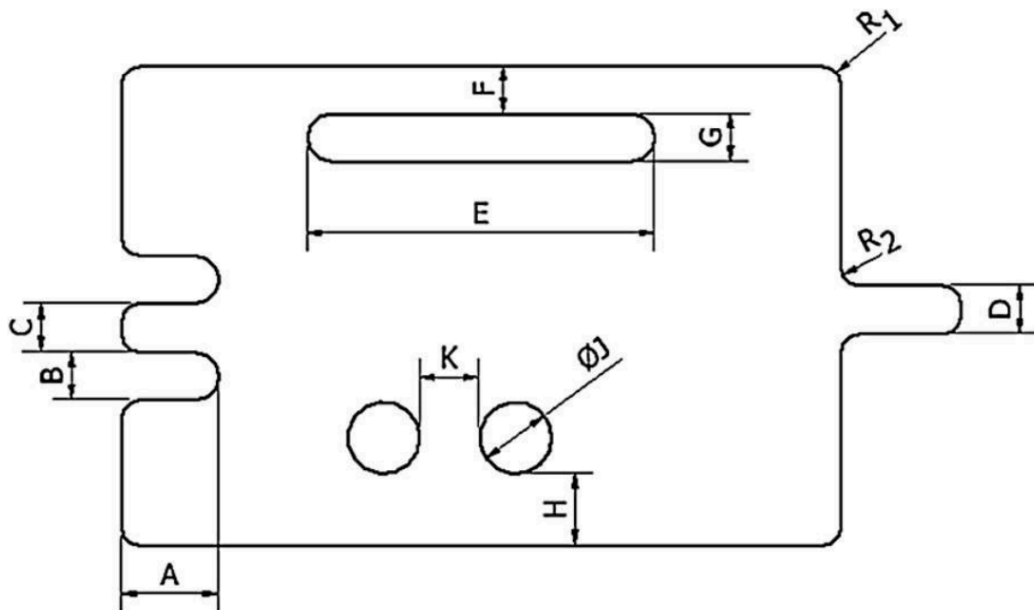


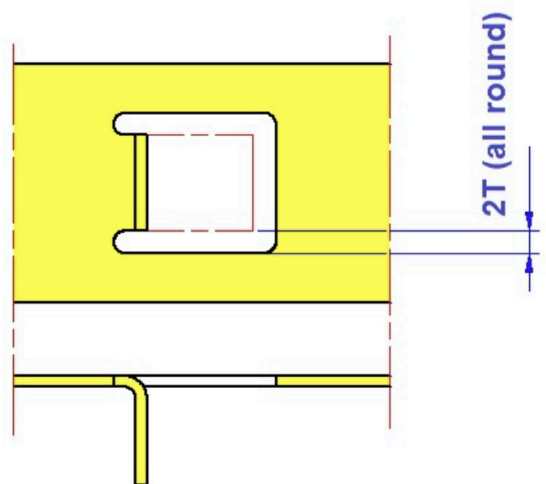
Bulging at Bends

compression at bends may cause slight bulging, reduced by cutting out material as relief:



Other Feature Clearances

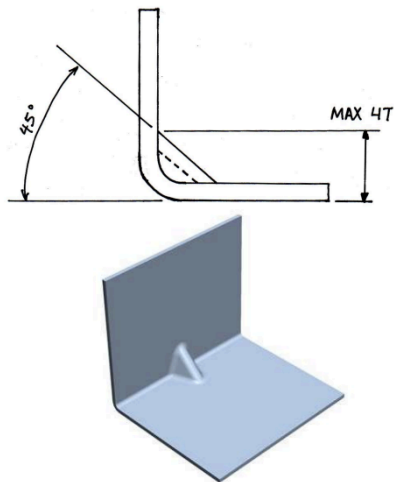




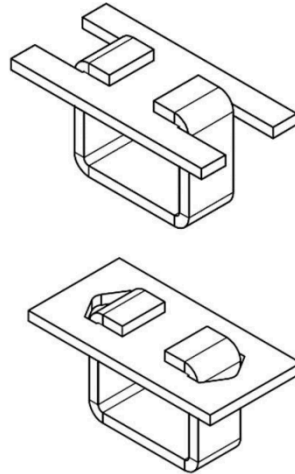
Feature Type	Dimension (assumed min unless otherwise)
Notch Depth A	max 6t when width 2t
Notch/Tab Width B, C and D	2t
Slot End-to-end length E	max 50mm when width=2t
Slot Edge distance to edge F	2t if width $\leq 10t$, else 4t
Slot width G	2t
hole edge distance, H and Hole to hole edge distance K	2t
hole diameter, J	1.2t
Radii, R1 and R2	0.5t or 4mm
Cut out for bent tab	2t
Gusset Height	4t
(covered earlier)	
flange length	2.5t+r
distance of hole/slot edge to bend	2.5t+r
notch bend relief width	2t
notch bend relief depth	t+r
circular bend relief radius	1.5r
extension tab bend relief	2t
corner relief	2t

Other stuff (sheet metal joints)

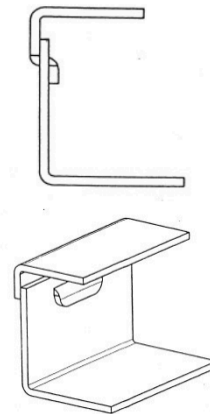
Guideline for Gusset



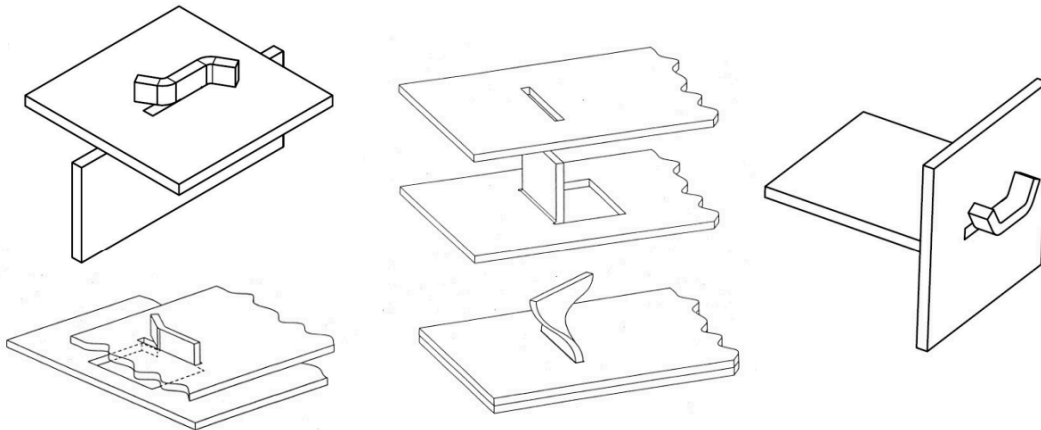
Bend Joint



Offset Bend Joint



Twist Tab Joints



Summary:

- Extremely sharp bends generally cannot be made. (Min Bend Radius)
- Bend Radius
 - Use a single bend radius for all bends to eliminate additional tooling or setups
- Bend Direction
 - Preferably bend across or perpendicular to rolling or grain direction for cold worked steel
- Hole near bend will be distorted. Place burrs on inside of bend
- Bend Relief or notches must be provided where necessary (Avoid tearing or fracture)
- Corner Relief for box shaped bending
- Developed bend length must be determined (Bend allowance)
- Accuracy of bend angle is affected by elastic recovery (Spring back)